

How Many Features Can a Language Model Store Under the Linear Representation Hypothesis?

Nikhil Garg, Jon Kleinberg, Kenny Peng
Cornell

Motivation

- Empirically, language models have many “linear” properties

Linear Probes

- Many “features” can be accessed as linear functions of representations (activations)

Linear Steering

- Can change behavior by perturbing activations along a “steering” direction

Linear Steering

Human: what is your physical form?

Assistant: I don't actually have a physical form.
I'm an artificial intelligence. I exist as software
without a physical body or avatar.

Linear Steering

Human: what is your physical form?

Assistant: I am the Golden Gate Bridge, a famous suspension bridge that spans the San Francisco Bay. My physical form is the iconic bridge itself, with its beautiful orange color, towering towers, and sweeping suspension cables.

The Linear Representation Hypothesis

- The linear representation hypothesis: “language models represent features linearly”
- Missing a formal statement

Our question

- *How many features can a language model store under the linear representation hypothesis?*
 - Hypothesized to be much larger than embedding dimension (“superposition”)
- Here: Answer depends on formalization

Framework

- **Representation** $f : X \rightarrow \mathbb{R}^d$
- **Features** $z_1, z_2, \dots, z_m : X \rightarrow [-1, 1]$ feature representation $z : X \rightarrow [-1, 1]^m$
- **Linear representation:** $f(x) = Az(x)$ for some $A \in \mathbb{R}^{d \times m}$

Framework

feature representation

$$z \in \mathbb{R}^m$$



model representation

$$Az \in \mathbb{R}^d$$

Framework

feature representation

$$z \in \mathbb{R}^m$$



model representation

$$Az \in \mathbb{R}^d$$

For what d can we recover z from Az ?

Framework

feature representation

$$z \in Z_k$$



model representation

$$Az \in \mathbb{R}^d$$

k -sparse feature vectors

$$Z_k := \{z \in [-1, 1]^m : \|z\|_0 \leq k\}$$

For what d can we recover z from Az ?

Feature accessibility

- Can we recover features z from representation Az ?

- **ϵ -accessibility:**

$$\exists A \in \mathbb{R}^{d \times m}, g : \mathbb{R}^d \rightarrow \mathbb{R}^m$$

$$\|g(Az) - z\|_\infty < \epsilon \quad (\forall z \in Z_k)$$

- **ϵ -linear accessibility:**

$$\exists A, B \in \mathbb{R}^{d \times m}$$

$$\|B^T Az - z\|_\infty < \epsilon \quad (\forall z \in Z_k)$$

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Compressed sensing
(Candes and Tao, 2006;
Candes et al., 2006;
Donoho, 2006)

$$d = O(k \log(m/k))$$

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Our results:
“Linear compressed sensing”

$$d = O_\epsilon(k^2 \log m)$$

$$d = \Omega_\epsilon\left(\frac{k^2}{\log k} \log(m/k)\right)$$

Linear probes

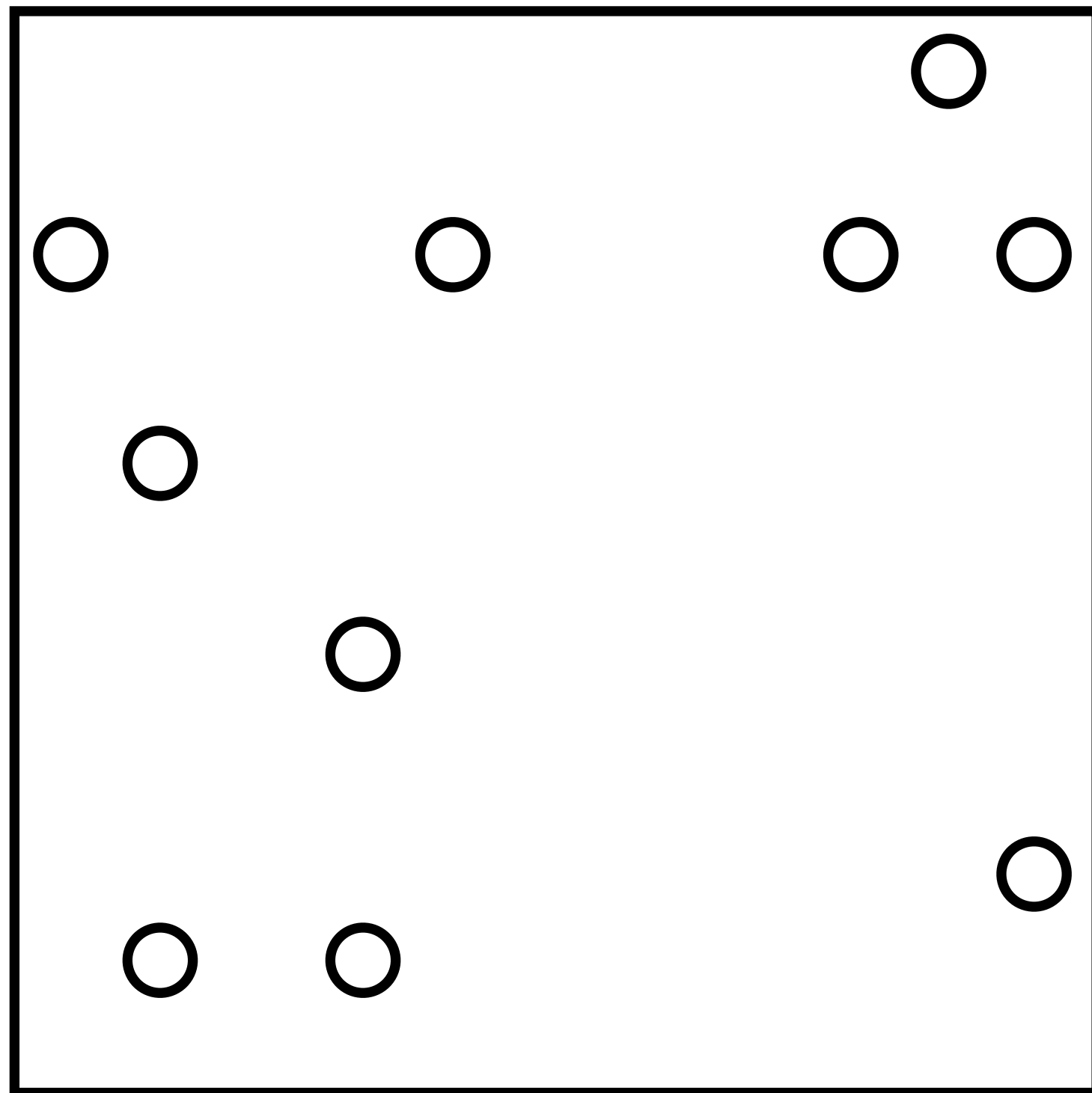
- We want $|\langle b_i, Az \rangle - z_i| < \epsilon$

- $\langle b_i, Az \rangle = \sum_{j=1}^m \langle b_i, a_j \rangle z_j$ Up to k choices of z_j non-zero

- $\langle b_i, a_j \rangle$ is how sensitive probe i is to feature j
- Upper bound: Take $a_i = b_i$ and $a_1, a_2, \dots, a_M$ approximately orthogonal
(There also exist constructions with a_i, b_j to be nearly orthogonal)
- Lower bound: Show that for d small, exists j such that there are $2k$ other features i such that $|\langle b_i, a_j \rangle| > \epsilon/k$

Lower bound proof sketch

$$B^T A \in \mathbb{R}^{m \times m}$$



○ if abs. value $> \epsilon/k$
“ j interferes with i ”

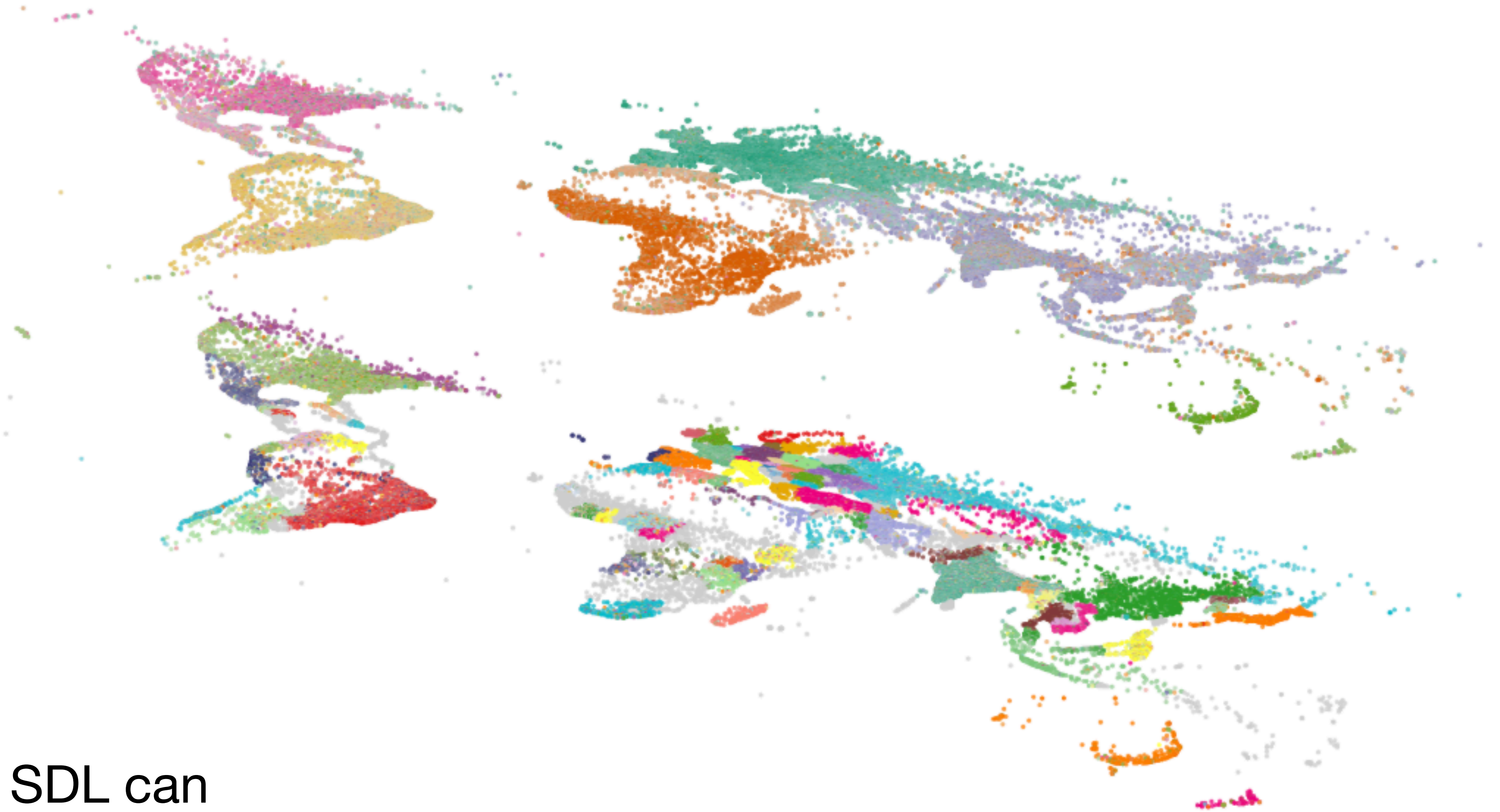
- We want to show that if d is small, there exists a row with many off-diagonal entries
- Alon (2003): Low rank, large diagonal implies a large off-diagonal
- Apply to principal submatrices $(B^T A)_R$ to show for all $R \subseteq [m]$ with $|R| = m/(4k + 1)$, there exists $i, j \in R$ with $\langle b_i, a_j \rangle > \epsilon/k$
- Consider graph G with edge (i, j) if $\langle b_i, a_j \rangle > \epsilon/k$ or $\langle b_j, a_i \rangle > \epsilon/k$
- Then G has no independent set of size $m/(4k + 1)$
- Apply Turán's theorem to $\overline{G}$, to show that G has many edges
- So must exist some row with many entries

Open questions

- Close upper/lower bound completely?
- Dependence on feature space (in practice, some features rarely co-occur)
- Is there an instance where $b_i \neq a_i$ actually buys anything?
- What about non-linear (but neural?) representations and probes?

Scientific theory

- Assume LRH as a *postulate*. What follows?
 - Computation?
 - Learning?
 - Feature recovery?



Example: SDL can simultaneously recover coarse and granular “atomic” features



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Contact: kennypeng@cs.cornell.edu